

▼ Data Analysis in Python Lab (1)

```
# installs the uncertainties package, and then imports the relevant commands.
!pip install --upgrade uncertainties
from uncertainties import ufloat
from uncertainties.umath import *
```

```
Collecting uncertainties
  Downloading uncertainties-3.2.3-py3-none-any.whl.metadata (7.0 kB)
  Downloading uncertainties-3.2.3-py3-none-any.whl (60 kB)
----- 60.1/60.1 kB 1.7 MB/s eta 0:00:00
Installing collected packages: uncertainties
Successfully installed uncertainties-3.2.3
```

▼ Part 2: Calculating Uncertainties

```
x = ufloat(10.34,0.04)
y = ufloat(7.66,0.03)
G = 6.67e-11
```

```
print((x**2 + y**2)**0.5)
print((4*x)/exp(y))
print((G*x)/y)
```

```
12.87+/-0.04
0.0195+/-0.0006
(9.00+/-0.05)e-11
```

▼ Part 3: Supernova Data

```
# imports the libraries needed to import data from google drive, read it in and plot it.
# grant permission to access your google drive.from google.colab import drive
drive.mount('/content/drive')
import pandas
import matplotlib.pyplot as plt
import numpy as np
```

Mounted at /content/drive

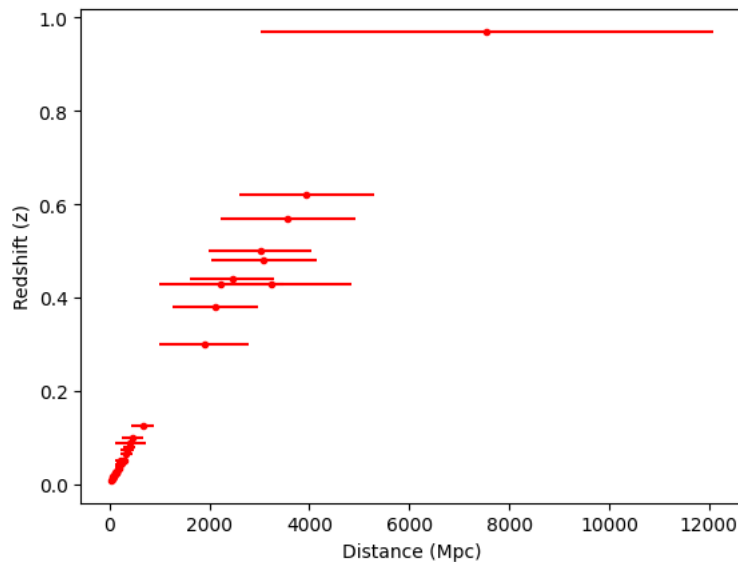
```
# reads data
data = pandas.read_excel("/content/drive/MyDrive/Python/Data/RawSupernovaData.xlsx")
data.head()
```

```
/usr/local/lib/python3.12/dist-packages/openpyxl/worksheet/_reader.py:329: UserWarning: Unknown extension is not supported
warn(msg)
```

	Supernova	Redshift	Distance	Uncertainty	
0	1995D	0.008334	36.140986	11.565116	
1	1995E	0.011746	55.718575	18.944315	
2	1992al	0.014057	66.988461	18.756769	
3	1995bd	0.015918	63.095734	22.714464	
4	1994S	0.016139	73.790423	20.661318	

Next steps: [Generate code with data](#) [New interactive sheet](#)

```
# plot the data read in above, and save the plot to your google drive.
# can change the formatting of the data points.
plt.errorbar(data.Distance, data.Redshift, xerr=data.Uncertainty, fmt="r.") # code to plot data with error bars
plt.xlabel("Distance (Mpc)") #can change the name of the columns you want to plot as needed
plt.ylabel("Redshift (z)")
plt.savefig("/content/drive/MyDrive/Python/errplot.png", dpi=300) # save plot as a png in drive
```



Part 4: Standard Deviations

```
# imports the libraries needed to import data from google drive, read it in and plot it.
# grant permission to access your google drive.from google.colab import drivefrom google.colab import drive
drive.mount('/content/drive')
import pandas
import matplotlib.pyplot as plt
import numpy as np
# installs the uncertainties package, and then imports the relevant commands.
!pip install --upgrade uncertainties
from uncertainties import ufloat
from uncertainties.umath import *
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True)

Collecting uncertainties

Downloading uncertainties-3.2.3-py3-none-any.whl.metadata (7.0 kB)

Downloading uncertainties-3.2.3-py3-none-any.whl (60 kB)

60.1/60.1 kB 2.2 MB/s eta 0:00:00

Installing collected packages: uncertainties

Successfully installed uncertainties-3.2.3

```
# reads data
data1 = pandas.read_csv("/content/drive/MyDrive/Python/Data/DataC.csv")
data1.head()
```

	Strength
0	12.6899
1	9.4204
2	10.9449
3	10.9476
4	8.3577

Next steps: [Generate code with data1](#) [New interactive sheet](#)

```
# reads data
data2 = pandas.read_csv("/content/drive/MyDrive/Python/Data/DataD.csv")
data2.head()
```

	Strength
0	10.1363
1	11.2901
2	9.4574
3	8.9479
4	11.5118

Next steps: [Generate code with data2](#) [New interactive sheet](#)

```
# code to compute the mean, median and standard deviation of data1
print("Number of data points = ", len(data1.Strength))
print("Mean=", np.mean(data1.Strength))
print("Median=", np.median(data1.Strength))
print("Standard Deviation=", np.std(data1.Strength))
```

```
Number of data points = 5000
Mean= 10.02093332
Median= 10.00715
Standard Deviation= 1.2320606607573252
```

```
# code to compute the mean, median and standard deviation of data2
print("Number of data points = ", len(data2.Strength))
print("Mean=", np.mean(data2.Strength))
print("Median=", np.median(data2.Strength))
print("Standard Deviation=", np.std(data2.Strength))
```

```
Number of data points = 5000
Mean= 10.09530266
Median= 10.0774
Standard Deviation= 1.0546041304484446
```

```
n_C = len(data1.Strength)
n_D = len(data2.Strength)
```

```
# code to compute the mean, uncertainty in mean, and difference in mean for data 1 & data2
mean_C = ufloat(np.mean(data1.Strength), np.std(data1.Strength) / n_C**0.5)
mean_D = ufloat(np.mean(data2.Strength), np.std(data2.Strength) / n_D**0.5)
diff = mean_C - mean_D
```

```
print("Alloy C mean strength=", mean_C)
print("Alloy D mean strength=", mean_D)
print("Difference (C - D)=", diff)
```

```
Alloy C mean strength= 10.021+/-0.017
Alloy D mean strength= 10.095+/-0.015
Difference (C - D)= -0.074+/-0.023
```

Part 4: Conclusion

The difference in mean strengths shows Alloy D is slightly stronger than Alloy C, not weaker.

it is observed that the new energy-efficient processs produced a marginally stronger alloy.